

MUHAMMAD RAFLI NUGRAHASYACH

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Business Statistics graduate (ITS) specializing in Machine Learning, NLP, and Deep Learning, with an international internship evaluating local LLMs on edge AI hardware (Universiti Teknologi PETRONAS, Malaysia), contribution to an institutionally funded research project (IDR 50M), and a co-authored paper published by Springer on Explainable AI for financial forecasting. BNSP-certified Junior/Associate Data Scientist experienced across the end-to-end data science workflow, from data processing to model development, evaluation, and dashboard delivery. Seeking to contribute technical expertise in Data Science, Machine Learning, and AI.

EDUCATION

- Bachelor of Applied Science (D4), Business Statistics | Institut Teknologi Sepuluh Nopember****Aug 2022 – Aug 2026**
- Focus: Machine Learning, Deep Learning, NLP, Time-Series Forecasting, Statistical Modeling
 - Relevant coursework: Text Mining, Deep Learning, Big Data Analytics
 - GPA: 3.51/4.00
- Data Science Bootcamp | Binar Academy (BINAR x Diskominfo Kota Tangerang Selatan)****Jun 2024 – Sep 2024**
- Best Performance Team; overall score 87/100.
 - Built a neural network / LSTM-based sentiment analysis model; handled data cleansing, model evaluation, and REST API development (Flask, Swagger UI).

PROFESSIONAL EXPERIENCE

- Data Scientist Intern | Center for Resilient Data Analytics and Computational Systems (CERDAS), Universiti Teknologi PETRONAS, Malaysia****Mar 2026 – Jul 2026**
- Stress-tested and evaluated the performance of local Large Language Models (LLMs) on an NVIDIA Jetson AGX Orin edge-AI device, scaling concurrent request loads (1, 4, and 8 users) to pinpoint the exact point of performance degradation, using model- and system-level performance metrics.
 - Built a pipeline to transform raw HTML data into structured JSON, and a Local Inference Layer to support on-device model processing.
 - Developed an analytics dashboard (React, Vite, Tailwind CSS) to visualize LLM testing results and performance analysis.
 - Designed scientific infographics to support the center's research publication output.
- Software Engineer | PT Automasi Visi Analitika Optima (Avalon AI Group)****Mar 2024 – Jul 2025**
- Developed an AI-powered Android learning application (Kotlin) integrating Speech-to-Text, Text-to-Speech, and Speech-to-Speech capabilities for voice-based interaction, as part of the Software Engineering and Operations division.
 - Organized and led AI-themed webinars and seminars attended by 100+ participants.
- Co-Founder & Chief Analytics Officer (CAO) | Upgrade UMKM, PT Digital Nusantara Innovations****Jan 2024 – Jan 2025**
- Co-founded Upgrade UMKM (5-person founding team) and served as Chief Analytics Officer, leading the development of BizTrack Monitor, an AI-powered business analytics platform for SMEs.
 - Built BizTrack Monitor's chat-based AI consultant feature, which analyzes uploaded business data (customer demographics, product sales) and generates natural-language executive reports with actionable business recommendations.
 - Designed data-collection schemas to aggregate sales data from multiple sources, improving data consistency for analytics.
 - Contributed to website development, SEO, and digital marketing to grow clients' online sales presence.
- Freelance Digital Business Analyst | CV Putra Terbaik****Jun 2023 – Present**
- Managed full-stack website development and technical SEO, and analyzed digital-marketing performance (keyword analysis, content tracking, Google Analytics) on a part-time / freelance basis, contributing to up to a 25% improvement in advertising effectiveness.
 - Designed marketing collateral (brochures, Instagram feed content, posters, and a 2026 company profile) to support the business's brand presence and sales channels.

RESEARCH EXPERIENCE & PUBLICATIONS

- Student Researcher | ITS Research Fund (Penelitian Kemitraan Tipe A) — PI: Dr. Noviyanti Santoso, Ph.D.****2025 – 2026**
- Contributed as a student researcher to an IDR 50,000,000 institutionally funded research project, "Multi-Modal Stock Price Prediction: A Comparative Study of Time-Series Forecasting, Sentiment Analysis, and Data Fusion in the Indonesian Market," led by Dr. Noviyanti Santoso.
 - Developed the LSTM with Attention Mechanism component as the basis for my undergraduate thesis (below), applying it to 8 LQ45-listed technology and financial stocks.
- Undergraduate Thesis: "Implementation of Explainable AI in Stock Price Forecasting of Technology and Financial Sectors in the LQ45 Index Using LSTM with an Attention Mechanism"****2025 – 2026**

- Built a multimodal data-fusion model combining historical price data with news and market sentiment (scraped from Stockbit), using IndoSBERT for semantic relevance and IndoRoBERTa for sentiment classification; applied SHAP for explainability.
- Found that the fusion model outperformed a pure technical baseline for high-liquidity stocks (e.g., BBCA), while the technical baseline was more effective for highly volatile stocks (e.g., GOTO), highlighting stock-specific model behavior.

Co-author, "Explainable AI and Gated Recurrent Units for Predicting Stock Movements in the IDX ESG Leaders Index" **2025 – 2026**

- The paper was presented at the 4th International Conference on Data Science and Emerging Technologies (DaSET 2025), UNITAR International University, and published in Springer Lecture Notes on Data Engineering and Communications Technologies (LNDECT), Vol. 302 (July 2026). DOI: 10.1007/978-3-032-23407-0_29.

Co-author, "Analisis Faktor-Faktor Tingkat Penggunaan E-Commerce di Setiap Provinsi Indonesia Menggunakan Analisis Faktor" — Socius: Jurnal Penelitian Ilmu-Ilmu Sosial **2024**

- Co-authored with F.M. Rizqi and S.P. Wulandari. Conducted statistical modeling (Factor Analysis / PCA) to identify key drivers of e-commerce adoption across Indonesian provinces. Vol. 2, No. 4, pp. 147–158, E-ISSN 3025-6704, DOI: 10.5281/zenodo.14187265.

Research Collaborator, "Smart Sentiment Monitoring System" — SVM-Based Digital Reputation Monitoring for Universities **2026**

- Cross-institution research collaboration with a researcher from Universitas Airlangga (UNAIR); contributed to data collection (web scraping from Google Maps and YouTube, 2,716 final data points across 2 universities) and text preprocessing for an SVM-based sentiment classification system. Project featured in Universitas Airlangga's Vokasi News (Jul 2026).

SELECTED ACHIEVEMENTS

12 national-level results in Statistics & Data Science competitions (2024–2026), including 3 first-place wins:

- 1st Place, Hasanuddin Statistics Event I, Universitas Hasanuddin (2026)
- 1st Place, National Statistics Challenge, Universitas Brawijaya (2025)
- 1st Place, Scientific Paper Competition – EXASTI 1.0, Universitas Negeri Jakarta (2025)
- 2nd Place, Airlangga Statistics Essay Competition (ASEC), Universitas Airlangga (2025)
- 2nd Place, Statistics Analysis Week – Jambore Statistika XIII, Universitas Mulawarman (2024)
- 2nd Place, Dokter Data, Universitas Diponegoro (2024)
- 3rd Place, Indonesian Vocational Olympiad (OLIVIA) X, FPTVI (2025)
- Top 5 Finalist, Jatim Datathon, East Java Communication & Informatics Office (2025)
- 2nd Best Team, ISE! Academy Python Programming for Data Science (Intermediate Class), ITS (2024)
- Top 5 Finalist, National Statistics Challenge, Universitas Brawijaya (2024)
- Top 7 Finalist, Airlangga Statistics Essay Competition (ASEC), Universitas Airlangga (2024)
- Top 10 Finalist, Data Competition – Information System Expo, Universitas Multimedia Nusantara (2024)

Research Grants & Business Funding

- Student Researcher, ITS Research Fund, Institut Teknologi Sepuluh Nopember (PI: Dr. Noviyanti Santoso) — IDR 50,000,000 institutionally funded project (2025)
- Growth Stage Funding, HETI Student Youngpreneurship (Upgrade UMKM), ITS & Partners — IDR 25,000,000 (2025)
- Early Stage Funding, ITS Youth Technopreneur (Upgrade UMKM), ITS & Partners — IDR 12,500,000 (2024)

ORGANIZATION & COMMITTEE

Senior Member, Data Scientist Division | Avalon AI Community **Nov 2024 – Nov 2025**

- Delivered Deep Learning and data visualization training materials to new members of the Data Scientist division.
- Participated in multiple external data science competitions as a community member, including a 3rd-place finish at the Indonesian Vocational Olympiad (OLIVIA) X (2025).

CERTIFICATIONS

- Junior/Associate Data Scientist — Badan Nasional Sertifikasi Profesi (BNSP), issued by Lembaga Sertifikasi Profesi ITS (LSP ITS); Competent, Sept 2025 (valid 3 years).
- Data Science Bootcamp Graduate — ISE! Academy Python Programming for Data Science, ITS (2024).
- Machine Learning & Data Visualization Track — Dicoding Indonesia (2023–2024): Python, SQL, Data Visualization, and Machine Learning for Beginners.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Transformers/Hugging Face, NLTK), SQL, R
Data Science / ML: Machine Learning, Deep Learning, NLP, Time-Series Forecasting, Explainable AI (SHAP), Statistical Modeling, Data Cleaning & Labeling, Web Scraping
Tools & Visualization: Tableau, Power BI, React, Git/GitHub, Google Cloud Platform (basic)

LANGUAGES

Indonesian: Native | English: CEFR B1 (Intermediate)